MAS1004 Data & AI, 2026 Fall

Assignment 2: Long report

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Leaderboard nickname: 허니머스타드. Current releases: Track A `track-a-1` (starter recipe) and Track B `track-b-1` (Track B starter baseline). Track A leaderboard score: Not observed. Track B leaderboard score: Not observed.

Track A

tag	what changed from the run before	validation
baseline, seed 0	Starter recipe; all 44,257 starter training photos; official budget of 249,984 photo exposures.	0.4967
baseline, seed 1	Same starter recipe and data as seed 0; seed changed only to measure run-to-run variation.	0.4948

The two baseline scores differed by 0.0019 (mean 0.4958). I used these two runs as a starting measurement and as an estimate of seed-to-seed variation. No cleaning, selection, augmentation, or architecture conclusion is drawn from the baseline alone.

The current Track A release is `track-a-1`, the unchanged starter recipe. Its leaderboard score was not observed, so no comparison between leaderboard and validation is claimed.

Decisions you made from photos

I did not copy class-only starter-photo contact sheets into this public repository. I inspected representative training sheets and the locally generated b1 mistake and suspect sheets for Busan and Osaka.

In the suspicious training images, I observed some isolated flowers, birds, food, museum objects, and indoor spaces without visible urban surroundings. In the error sheets, Osaka street and traditional-building scenes were often predicted as Tokyo, while Busan scenes were often predicted as Seoul. The saved Track B b1 confusion matrix recorded 785 Osaka validation photos predicted as Tokyo and 410 Busan validation photos predicted as Seoul.

I wrote the following prospective rule before acting: exclude only training photographs that show isolated objects, food, animals, or other close-up subjects without recognizable urban or location-specific context. This rule is **untested**: no photo was removed, no selection file was created, and no follow-up model was trained before the deadline.

The relevant requests were: "Some suspicious training images contain almost no useful information about their city" and "Exclude only training photographs that show isolated objects, food, animals, or other close-up subjects without any recognizable urban or location-specific context." I also specified that the model architecture, augmentation, random seed, and official training budget should stay unchanged for a future comparison. No action was taken on that proposal.

Saved observation and decision record

```
Inspection observations and proposed rule, supplied by the student on 9 November 2026:

1. Some suspicious training images contain almost no useful information about their city. For
↪  example, there are close-up photos of flowers, birds, food, museum objects, and indoor spaces
↪  without visible urban surroundings.

2. The model frequently confuses visually similar cities. Osaka streets and traditional buildings
↪  are often predicted as Tokyo, while Busan scenes are often predicted as Seoul.
```

```
Proposed controlled experiment:

Exclude only training photographs that show isolated objects, food, animals, or other close-up
↪  subjects without any recognizable urban or location-specific context.

Keep the model architecture, augmentation, random seed, and official training budget unchanged.

Status: UNTESTED. The student instructed that no images be removed and no additional training be
↪  started because the deadline was extremely close.
```

Track B

tag	what changed from the run before	validation
b1	Starter Track B baseline: ResNet18 at 224 pixels, ImageNet-1K V1 starting weights, `data/starter/train`, 4 epochs.	0.4961

The b1 submission passed the local format and limit checker: 11.17 million parameters, 1.81 billion multiply-adds per 224-by-224 photo, and a 44.7 MB ONNX file. This established a valid starter Track B submission before attempting a controlled improvement. No changed Track B model was trained.

No extra photos were used in b1; it trained on `data/starter/train`.

The b1 validation score was 0.4961. Its leaderboard score was not observed, so no comparison between leaderboard and validation is claimed.

Your submission, as the grader sees it

```
D:\Data&AI\mas1004-assignment2\submission\model.onnx: 44.7 MB, 11.17 million parameters, 1.81
↪  billion multiply-adds per 224 x 224 photo
  limits: 12 million parameters, 2.0 billion multiply-adds
  format and limits: OK

D:\Data&AI\mas1004-assignment2\data\starter\val: score 0.496 on 12077 photos (278s)
  seoul 0.556, busan 0.388, tokyo 0.611, osaka 0.384, taipei 0.541
  confusion (rows: true city, columns: answer):
    seoul    1355   173   438   176   293
    busan     410   539   219    75   147
    tokyo     559    89  2371   543   319
    osaka     277    61   785   817   188
    taipei    359    72   437   161  1214
```

What made the difference

No changed cleaning, augmentation, or additional-data experiment was completed before the deadline. Therefore, this report does not claim that any change helped or hurt.

With one more week, I would test the written close-up/isolated-subject exclusion rule as one controlled Track A change, keeping the architecture, augmentation, seed comparison, and official budget fixed. I would then compare its validation score with the baseline and inspect the new errors before making another change.

All runs

```
finished,track,tag,seed,official_budget,budget,val_score,seoul,busan,tokyo,osaka,taipei,train_seco⌋
↪  nds,device,selection,recipe_sha,selection_sha,commit
2026-11-09T13:23:32,A,baseline,0,True,249984,0.4967,0.5696,0.3777,0.6426,0.3459,0.5479,4021.0,cuda⌋
↪  ,all training
↪  photos,17ac485223fa,,84b5b12
2026-11-09T14:03:26,A,baseline,1,True,249984,0.4948,0.5696,0.3935,0.6256,0.3647,0.5207,2284.3,cuda⌋
↪  ,all training
↪  photos,17ac485223fa,,84b5b12
2026-11-09T15:40:22,B,b1,,,176896,0.4961,0.5565,0.3878,0.6109,0.3839,0.5412,1667.0,cuda,resnet18
↪  224px; data/starter/train,,,3028823
```

Lab notebook

```
# Log

One entry for each recipe, selection or Track B model you train, newest at
the bottom. The numbers come from `experiments.csv`. "Why" and "what it means"
are in your own words: your agent asks you for them and does not make them up.
End the log with your plan for what to try next.

<!-- The form of an entry:

## <date> <tag> (Track A or B)
- changed: what is different from the run before
- why: your reason
- validation: 0.000 (before: 0.000)
- what it means: what you think the result says
-->

## 2026-11-09 baseline (Track A)
- changed: Starter `track_a/recipe.py`, all 44,257 starter training photos, and the official
↪  249,984-photo training budget. Two runs were made only to measure seed-to-seed variation.
- why: Establish a measured starting point and estimate seed-to-seed variation before testing any
↪  individual change.
- validation: seed 0 = 0.4967; seed 1 = 0.4948 (difference = 0.0019; mean = 0.4958).
- what it means: The observed score difference between these two baseline seeds was 0.0019. No
↪  cleaning, augmentation, selection, or architecture conclusion is drawn from this baseline
↪  alone.

## 2026-11-09 b1 (Track B)
- changed: Starter Track B configuration: ResNet18 at 224 pixels, ImageNet-1K V1 starting weights,
↪  `data/starter/train`, and 4 epochs. The submitted model passed `check_submission.py` format
↪  and limit checks.
- why: Establish the supplied Track B starting point and produce a valid submission before
↪  attempting a controlled improvement.
- validation: 0.4961 (Seoul 0.5565, Busan 0.3878, Tokyo 0.6109, Osaka 0.3839, Taipei 0.5412).
- what it means: This is the validated starter Track B reference model. It is not evidence that a
↪  later data-cleaning hypothesis has helped, because no changed Track B model was trained.

## 2026-11-09 inspection and untested proposed next step
- inspected: Representative training sheets and the Track B b1 mistake/suspect sheets for Busan
↪  and Osaka. The Track A ONNX files were not returned from Colab, so the mistake and suspect
↪  sheets are explicitly from the local Track B b1 submission model, not relabelled as Track A
↪  baseline sheets.
- observations: Some suspicious training images showed isolated flowers, birds, food, museum
↪  objects, or indoor spaces without visible urban surroundings. Osaka street and
↪  traditional-building scenes were often predicted as Tokyo, while Busan scenes were often
↪  predicted as Seoul.
- proposed controlled rule: Exclude only training photographs that show isolated objects, food,
↪  animals, or other close-up subjects without recognizable urban or location-specific context.
```

```
- status: UNTESTED. No photo was removed, no `selection.csv` or `dropped.csv` was created, no
↪  augmentation or architecture setting was changed, and no additional training was started
↪  before the deadline.
- next action if more time were available: Apply this written rule only, keep the model
↪  architecture, augmentation, random seed, and official training budget unchanged, then compare
↪  the new result fairly against the Track A baseline.
```